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Customization of online advertising: The role of intrusiveness

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Abstract This study examines the trade-off between tailoring an ad to an individual consumer's needs and the demand for personal data required to do so, which is inherent in targeted online advertising. Two scenario-based studies in two different industries reveal the extent to which using personal information with greater distinctiveness affects consumers' sense of intrusiveness and purchase intentions. We also manipulate ad fit to each consumer's needs and the presence of a discount to investigate whether feelings of intrusiveness mitigate the potential positive effects of fit and discount. Higher degrees of personalization, such as adding personal identification or transaction information to browsing data, increase feelings of intrusiveness, and negatively affect purchase intentions. These negative effects are not compensated for by offering discounts but can be partly mitigated by presenting an ad with a high fit to consumers' current needs. The positive effect of tailoring the ad is weakened at higher levels of intrusiveness though. Furthermore, high fit may lead to not only higher purchase intentions but also higher perceived intrusiveness levels. Presenting a consumer with a customized ad can thus be a double-edged sword, leading to higher purchase intentions but also greater perceived intrusiveness, which then negatively affects purchase intentions.

Keywords Intrusiveness · Online advertising · Customization · Personalization · Targeting

The use of customized advertising on the Internet based on personal information about a customer, beyond just information gathered through his or her browsing history, is becoming more and more common, though it also has come under increased public scrutiny. For example, recent plans by the two largest credit card networks, Visa and Mastercard, to use information about their customers' shopping

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behavior to target them with customized online advertising prompted worried reactions (Steel 2011). From a consumer perspective, customized advertising has the advantage of greater relevance and fit, such that it offers the right product or service at the right moment and makes the information search process more efficient by relieving consumers of the need to search further (Chellappa and Sin 2005; Okazaki et al. 2009; Tam and Ho 2006). Customized advertising research shows that offers that provide a high fit with a consumer's preferences may increase his or her purchase intentions (Franke et al. 2009; Goldfarb and Tucker 2011), whereas low fit offers induce irritation (Thota and Biswas 2009) or even reactance if the recommendations are contrary to the consumer's choice tendencies (Fitzsimons and Lehmann 2004).

The benefits of customized ads also come with some mental or psychological costs. Customized ads require customer insights and the use of personal information, which may seem too personal, because it might demand an unwanted level of knowledge of the consumer's preferences and behavior (Tucker 2011; White et al. 2008). Although the fit of the customized ad may increase the ad's relevance, and thus result in positive behavioral effects, the use of more personal information may induce feelings of intrusiveness that interfere with the consumer's cognitive processing and interrupt goal pursuit (Li et al. 2002), such that it prevents the consumer from taking notice of the ad contents (Morimoto and Chang 2006). Intrusive ads also may be perceived as annoying and result in reactance, such that consumers behave in the opposite way to the one intended by the advertiser (Clee and Wicklund 1980; Ying et al. 2009). The effect of potential benefits therefore may be smaller or nonexistent if customized ads seem intrusive. This trade-off is intriguing; in many cases, the benefits and psychological costs of customized advertising go hand in hand, such that data usage is a prerequisite of providing consumers with relevant information. This dilemma raises a key question: can the effectiveness of providing consumers with ads with high fit be attenuated by feelings of intrusiveness, triggered by personalizing the message with information necessary to create these ads?

Prior literature offers some evidence that a consumer's reaction to a customized message depends on both the degree of customization and the potential benefits of the offer. For example, White et al. (2008) present respondents with messages that differ in their amount and distinctiveness of personal information, the extent to which the use of information is justified, and the perceived utility of the message. They find that perceived utility attenuates negative reactions to customized messages, though this effect is weaker in the presence of highly distinctive personal information. Goldfarb and Tucker (2011) investigate the joint effects of contextual targeting—that is, presenting an ad on a website that matches the topic of the ad—and ad obtrusiveness, such as including visual and audio features that make it harder to ignore the ad. Purchase intentions increase in response to online advertising if the ad matches the content of the website and to obtrusive ads, if either is used in isolation, but not if the techniques are combined. Although that study does not personalize the ads at different levels of distinctiveness, it shows that the effectiveness of presenting relevant content can diminish due to feelings of intrusiveness, induced by actions that a consumer might perceive as an invasion of her or his personal space. Tam and Ho (2006) find that both a better fit of the offer and the use of more personal information (e.g., name) have positive effects on choice behavior.

Studies related to the customization of online ads thus far have failed to address two relevant issues though. First, the explicit trade-off between message relevance and feelings of intrusiveness triggered by personalization has not been examined. As online advertising grows more popular (Boris 2012) and technology increasingly facilitates sophisticated ways to customize ads, enhancing knowledge in this field is essential. We therefore study the effect of the use of personal information on intrusiveness, together with the moderating role of intrusiveness on the effect of ad fit. Second, previous studies that address both positive and negative effects of customized ads use browsing data (Goldfarb and Tucker 2011; Tucker 2011) or demographic data (White et al. 2008) to personalize the message. However, as our preceding example illustrates, using transaction information for targeting is becoming more common in marketing practice. Furthermore, transaction information enables the advertiser to match the ad's content with the specific needs of an individual consumer. We therefore use information about previous transactions to personalize ads. Considering the importance of prices for ensuring the effectiveness of propositions in general (Compeau and Grewal 1998) and the increasing popularity of discounts as a tool to support targeted online advertising (Helft and Vega 2010), we extend our framework to examine the impact of discounts in an online targeting context.

The purposes of our research thus are to investigate (1) the extent to which the use of different types of information for personalizing customized ads triggers feelings of intrusiveness, such that we explicitly include the use of transactional data, (2) whether feelings of intrusiveness affect purchase intentions, (3) whether feelings of intrusiveness lead to less positive reactions to customized ads with high fit, and (4) the extent to which feelings of intrusiveness might attenuate the potential positive effect of a discount. We perform our study in a customer relationship setting, in which the advertiser possesses data about customers' backgrounds and transactions, and customers generally trust the advertiser because of the nature of the business they already have conducted with the company. Our results show that information with greater distinctiveness (i.e., personal or transaction information, versus browsing data) increases feelings of intrusiveness, which negatively affect purchase intentions. Fitting the ad to the consumer's needs leads to higher purchase intentions, but this effect gets weakened by the ad's intrusiveness. Furthermore, one of our studies shows that high fit may lead to both higher purchase intentions but also higher levels of perceived intrusiveness. The customization of online advertising therefore offers a double-edged sword: It may heighten purchase intentions, but it also enhances perceptions of intrusiveness, which negatively affect purchase intentions. Our results imply that even in customer relationships, companies should account for the level of intrusiveness that their customized ads invoke. Surprisingly, we did not find any effects of discounts on purchase intentions. In the following sections, we present our conceptual model and theory, along with the results of the two studies we performed, and our conclusions.

1 Conceptual model

Figure 1 contains our conceptual model. The core variable is intrusiveness, a construct rooted in psychology. It encompasses "creating an imbalance between closeness and autonomy" (Lavy et al. 2009, p. 990), where closeness refers to the degree of relatedness

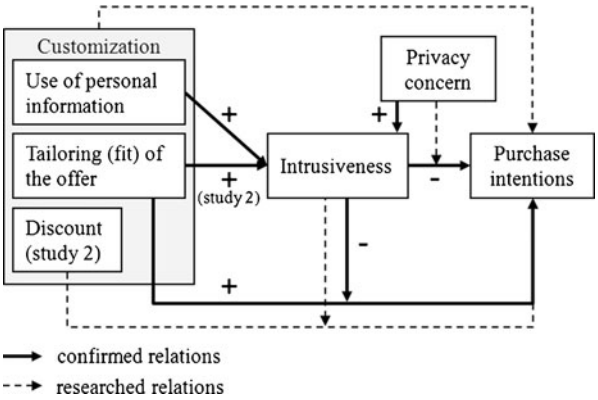


Fig. 1 Conceptual model and results

and interdependence between two parties, and autonomy is the degree to which personal identity can be preserved. In an advertising context, a sense of intrusiveness is "a psychological reaction to ads that interfere with a consumer's ongoing cognitive processing" (Li et al. 2002, p. 39). We expect intrusiveness to be influenced by the degree of customization of the targeted ad, which consists of two components: personalization and fit. Following White et al. (2008), we apply the concept of personalization to differentiate messages for different recipients, such that the type of data used for personalization determines the degree to which the personal information uniquely identifies or characterizes its recipient. Messages based on browsing data may be applicable to a large number of consumers and do not uniquely identify a recipient. Adding information such as names and past transactions to a message greatly increases its distinctiveness, and therefore its intrusiveness. We distinguish in our research between ads based on browsing data only, ads based on browsing data that are personalized by using the web visitor's name, and ads based on the consumer's actual browsing behavior linked with his or her transaction history. We expect intrusiveness to have a negative impact on purchase intentions (White et al. 2008). Our choice of purchase intentions as the dependent variable is in line with previous research in this field (e.g., Chellappa and Sin 2005; Goldfarb and Tucker 2011).

As a benefit of customization, an ad with high fit is tailored to the needs of the consumer and presents relevant information; high fit therefore should increase a consumer's purchase intentions (Franke et al. 2009; Tam and Ho 2006). Yet a message with high fit may also reveal that the supplier has used information about the consumer, which this consumer may perceive as a loss of control (Edwards et al. 2002) or as intrusive. Considering extant evidence that the positive effect of the potential benefits of customized advertising can be offset by overly distinctive information (White et al. 2008) or an ad that is too prominent (Goldfarb and Tucker 2011), we expect perceived intrusiveness to reduce the positive effects of presenting an ad with a high fit (study 1). In study 2, we also investigate whether feelings of intrusiveness affect a consumer's reaction to promotional prices. Prior research cites the effects of certain consumer traits, such as privacy concerns, on emotional reactions (e.g., Okazaki et al. 2009) and purchase intentions (Goldfarb and Tucker 2011). We therefore include privacy concerns in our model and expect consumers with higher levels of this trait to react more

negatively to intrusive ads than do consumers who are less concerned about their privacy.

2 Study 1

2.1 Method

In a qualitative prestudy, we verified our operationalization of the customization of online advertising through 12 in-depth interviews with managers who used online targeting in their customer strategies, as well as two focus group sessions with consumers who evaluated their experiences with and attitudes toward customized advertising. Next, 233 participants on a consumer panel in the Netherlands undertook a 3 (personalization—ad based on browsing data only, ad based on browsing data and the person's name, and ad based on browsing data and information about previous transactions)×2 (ad fit—low vs. high) between-subjects, factorial, online experiment. On average, 39 respondents considered each of the six scenarios, through a random assignment. Participants started by reading the text on a computer screen; the text asked them to imagine themselves as a person called Henny de Vries (a name that could be male or female in the Netherlands), who is a customer of VCO Bank (fictitious name), uses Internet banking to deal with his/her financial affairs, and has just gotten a raise in pay. The text also states that the raise has prompted Henny to begin shopping for a new house, such that he/she checks the VCO site for information about mortgages; on the next day, Henny visits the same site to make some bill payments. The next screen of the experimental task presented the fictitious website of VCO Bank, with a banner that reflects one of the six scenarios, offering either a mortgage (high fit) or a credit card (low fit). (See Appendix for an example of stimuli used in the study). The degree of personalization was manipulated through the use of a general banner ("Buy a … at VCO Bank now"), a personalized banner using the person's name, or a banner that reveals details of the person's transaction history (e.g., current mortgage for the mortgage scenarios or pay raise for the credit card scenarios). Next, we measured intrusiveness, purchase intentions, and privacy concerns using existing scales (see Table 1). For intrusiveness, we used ten items adapted from Mooradian (1996) and Edwards et al. (2002) (Cronbach's alpha=0.92). We used three items from Grewal et al. (1998) to measure purchase intentions (Cronbach's alpha=0.85) and three items adapted from Culnan (1993) to measure privacy concerns (Cronbach's alpha=0.77). We pretested the questionnaire in the focus groups. The means of privacy concerns did not differ significantly across experimental conditions (p (ANOVA)>0.1). We also mean centered privacy concerns to alleviate multicollinearity.

We used a system of equations to test our conceptual model, which shows that we expect impacts of the different types of personalization (browsing data is the base category), the fit of the ad, and privacy concerns on perceived ad intrusiveness (Eq. 1). A consumer's purchase intentions should be negatively affected by the perceived intrusiveness of the offer and positively affected by the offer's fit; the positive effect of fit should be weaker for intrusive offers (Eq. 2). We further expect the negative effect of intrusiveness on purchase

Table 1 Scale items

Construct	Items	Source	CA study 1	CA study 2
Intrusiveness	I think this offer is disturbing	Edwards et al. (2002) and Mooradian (1996)	0.92	0.93
	I think this offer is alarming			
	I think this offer is obtrusive			
	I think this offer is irritating			
	I think this offer is annoying			
	I think this offer is uncomfortable			
	I think it is uncomfortable that personal information is used in this offer			
	The supplier knows a lot about me			
	This offer gives me an uneasy feeling			
	This offer gives me an unsafe feeling			
Purchase intentions	The likelihood of purchasing this product is large	Grewal et al. (1998)	0.85	0.86
	The probability that I would consider buying the product is large			
	If I am going to buy a …, the probability of buying this model is large			
Privacy concern	I am concerned about threats to my personal privacy	Culnan (1993)	0.77	0.77
	Consumers have lost all control over how personal information is used			
	Citizens begin surrendering their privacy the day they open their first checking account			

intentions to depend on the respondent's level of privacy concern, and we allow for a direct effect of the different types of personalization on purchase intentions. Thus,

$$\text{intrusiveness} = \alpha_1 + \beta_1 \cdot \text{name} + \gamma_1 \cdot \text{transaction} + \delta_1 \cdot \text{fit} + \phi_1 \cdot \text{privacy} + \varepsilon_1 \quad (1)$$

$$\begin{aligned}\text{intention} = {} & \alpha_2 + \beta_2 \cdot \text{name} + \gamma_2 \cdot \text{transaction} + \delta_2 \cdot \text{fit} + \psi_2 \cdot \text{intrusiveness} \\ & + \lambda_2 \cdot \text{intrusiveness} \times \text{fit} + \phi_2 \cdot \text{privacy} + \theta_2 \cdot \text{intrusiveness} \times \text{privacy} + \varepsilon_2\end{aligned} \quad (2)$$

where

intrusiveness = perceived intrusiveness of the ad
name = dummy variable, equal to 1 if the ad is personalized with the person's name, and 0 if otherwise
transaction = dummy variable, equal to 1 if transaction information is used, and 0 if otherwise
fit = dummy variable, equal to 1 if ad has a high fit, and 0 if otherwise

privacy = respondent's level of privacy concern; and
intention = purchase intentions.

For the model estimation, we used a seemingly unrelated regressions (SUR) estimator to account for the correlation of the disturbances (e.g., Greene 2008). A Breusch–Pagan test revealed that the residuals were not independent ($p<0.01$), so using the SUR estimator is required.

2.2 Results

From Table 2, we determine that perceived intrusiveness is greater when a message is personalized with the name ($\beta=0.51$, $p<0.05$) or uses transaction information in addition to browsing data ($\beta=0.58$, $p<0.01$). Respondents with higher levels of privacy concerns perceive the ad as more intrusive ($\beta=0.40$, $p<0.01$), but high fit does not affect intrusiveness ($\beta=-0.16$, $p>0.1$). In line with our expectations, respondents are less likely to purchase intrusive offers ($\beta=-0.23$, $p<0.01$) and more likely to purchase offers with high fit ($\beta=1.67$, $p<0.01$), but this effect is weaker if intrusiveness is high ($\beta=-0.28$, $p<0.01$). Privacy concerns and the different degrees of personalization do not significantly affect purchase intentions ($p>0.1$). We performed two additional analyses to rule out alternative explanations. First, we examined whether the effect of varying degrees of personalization on intrusiveness differed according to the fit of the ad. To this end, we included interaction effects between the dummies signifying the varying degrees of personalization and fit; they all failed to reach a satisfactory level of significance. Second, we investigated whether privacy concern moderated the effect of the use of different types of personalization on intrusiveness. It did not.

Table 2 Results of study 1

Independent variables	Dependent variable[a]	
	Intrusiveness	Purchase intentions
	Coefficient	Coefficient
Personalization: use of name	0.51**	0.04
Personalization: use of transaction information	0.58***	0.08
High fit	−0.16	1.67***
Intrusiveness		−0.23**
Intrusiveness×high fit		−0.28***
Privacy concern	0.40***	0.07
Intrusiveness×privacy concern		−0.05
Constant	4.33***	3.20***
R^2	0.17	0.27
N	233	233

**$p<0.05$; ***$p<0.01$ (two-sided test)

[a] Results obtained using a seemingly unrelated regressions estimator

3 Study 2

3.1 Method

In a second study, we replicated our results from study 1 in the telecommunications industry. Furthermore, noting the importance of prices for promotion effectiveness (Compeau and Grewal 1998), we investigated whether the effectiveness of granting a discount depended on the use of more personal information in a customized ad. We chose the telecommunication industry for this study, because according to our in-depth interviews with managers, discounts are more credible in the telecom than in banking industries. To this end, 467 respondents to a consumer panel, who did not take part in study 1, participated in a 3 (personalization—ad based on browsing data only, ad based on browsing data personalized using the person's name, and ad based on browsing data and information about previous transactions)×2 (ad fit—low vs. high)×2 (discount—10 % or no discount) between-subjects factorial experiment. On average, 39 randomly assigned respondents considered each of the 12 scenarios, such that they again played the role of Henny de Vries who was a customer of the telecom provider TeleNow (fictitious name). The respondents were to imagine that their telephone subscription would expire in three months and that they had been searching for information about new subscriptions on TeleNow's website. Next, the respondents reviewed the fictitious website of TeleNow, which featured a banner describing one of the 12 scenarios. (See Appendix for an example of stimuli used in the study). In addition to offering either a phone subscription (high fit) or an Internet subscription (low fit), we adopted the same procedure as in study 1 and used a general banner, a banner addressing the web visitor by name, or a banner revealing the customer's calling behavior to manipulate the different types of personalization. The 10 % discount was available in only some scenarios. We used the same scales to measure intrusiveness (Cronbach's alpha=0.93), purchase intentions (Cronbach's alpha=0.86), and privacy concerns (Cronbach's alpha=0.77) as in study 1. We also again compared the means of privacy concern across the different experimental conditions and did not find any significant differences (p (ANOVA)>0.1).

3.2 Results

As the results in Table 3 show, study 2 replicates the majority of the findings from study 1. Personalization of the ad with the name (β=0.49, p<0.01) and using information about transactions (β=0.66, p<0.01) increase the perceived intrusiveness of the ad. Higher privacy concerns lead to perceptions of the ad as more intrusive (β=0.40, p<0.01). However, unlike our findings in study 1, we determine that an ad with high fit is perceived as more intrusive (β=0.15, p<0.05). As in study 1, purchase intentions are lower if the ad is perceived as intrusive (β=−0.13, p<0.1) but higher if fit is high (β=0.40, p<0.05). The positive effect of higher fit gets partly offset if the ad is intrusive (β=−0.09, p<0.05). Although higher privacy concerns lead to lower purchase intentions (β=−0.35, p<0.01), privacy concerns do not moderate the link between intrusiveness and purchase intentions. Surprisingly, we also do not find any significant impact of a discount on purchase intentions. Again, we performed two additional analyses: We examined whether the effect of the use of different types of

Table 3 Results of study 2

Independent variables	Dependent variable[a]	
	Intrusiveness	Purchase intentions
	Coefficient	Coefficient
Personalization: use of name	0.49***	−0.05
Personalization: use of transaction information	0.66***	−0.10
High fit	0.15**	0.40**
Discount		0.13
Intrusiveness		−0.13*
Intrusiveness × high fit		−0.09**
Intrusiveness × discount		−0.01
Privacy concern	0.40***	−0.35***
Intrusiveness × privacy concern		0.04
Constant	3.79***	3.16***
R^2	0.20	0.16
N	467	467

*p<0.1; **p<0.05; ***p<0.01 (two-sided test)

[a] Results obtained using a seemingly unrelated regressions estimator

personalization on intrusiveness differs depending on ad fit, and we checked for moderating effects of privacy concern on intrusiveness. In neither case did we find any significant effects.

4 Conclusions

With this research, we have sought to highlight the role of perceived intrusiveness in customized online advertising and consumers' purchase intentions. Toward this end, we conducted two scenario-based studies in two industries and investigated the extent to which using personal information with increasing distinctiveness, such as by enriching browsing data with personal identification or transaction data, affects the sense of intrusiveness and purchase intentions. Although previous literature has examined the use of demographic information and its potential to prompt personalization reactance (White et al. 2008), we offer the first study that also investigates the effects of using transaction data for targeting purposes, a practice that has become more common in the Internet advertising era. Unlike previous research (Goldfarb and Tucker 2011; White et al. 2008), we also manipulate the fit of the presented advertisement with the consumer's needs and investigate whether strong feelings of intrusiveness lessen the potential positive effect of providing an ad with high fit. In so doing, we explicitly examine the trade-off between tailoring an ad to a consumer's needs and using personal data to do so, which is an inherent element of targeted online advertising. Finally, in an extension of previous literature, we examine whether

the effectiveness of discounts is affected by personalization or feelings of intrusiveness.

We find that higher degrees of personalization, such as adding personal identification or transaction information to an ad, beyond just using browsing data, increase feelings of intrusiveness and negatively affect purchase intentions. These negative effects of personalization can be partly compensated for by presenting consumers an ad that effectively fits their current needs. Yet the positive effects of tailoring the ad to consumers' needs must be regarded with caution. First, the positive effect is weaker when the ad is more intrusive. Second, in our telecom sector study 2, high fit led to not only higher purchase intentions but also to higher levels of perceived intrusiveness. This result is intriguing; attempts to serve consumers better by providing messages that are relevant to them can backfire. Perhaps consumers interpret high ad fit as yet another piece of evidence that the advertiser is using their personal information, which they may perceive as a loss of control (Edwards et al. 2002). But why did fit lead to higher perceived intrusiveness only in the telecommunication sample? We propose that consumers in general may have a higher degree of trust in their banks than in their telecommunication services providers, which according to Simonson (2005) would increase the attractiveness of a customized offer from the bank. Unfortunately, we cannot confirm that claim conclusively here; instead, we suggest that the effects of customized Internet advertising likely differ across industries.

In Fig. 2. we graphically display the net effects of using varying degrees of personal information for ads with a high or low fit on purchase intentions in both industries. In the financial services study, purchase intentions are always higher for ads with high fit. However, due to the effect of intrusiveness, the advantage of fit diminishes when the ad uses the name of the web visitor or includes transaction information. Because in the telecommunication example, high fit ads appear more intrusive, presenting a high fit ad to a customer does not put a company at an advantage. This result is in line with White et al.'s (2008) finding that messages using more distinctive personal information do not result in higher clickthrough intentions.

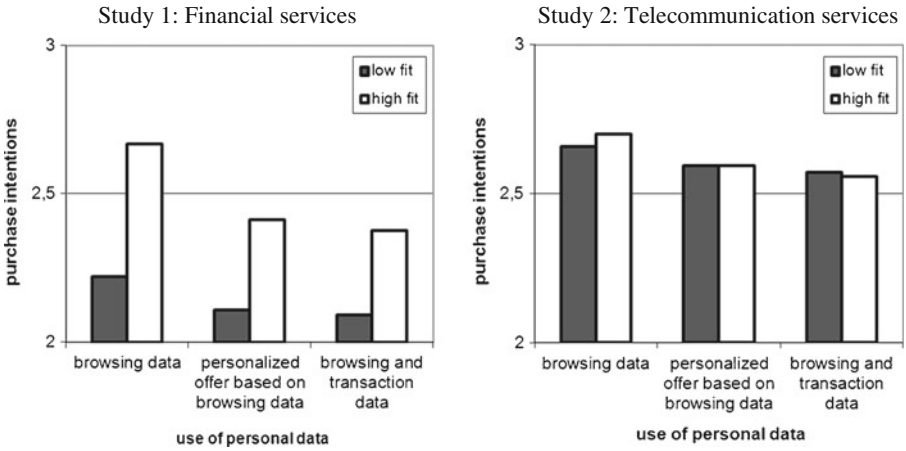


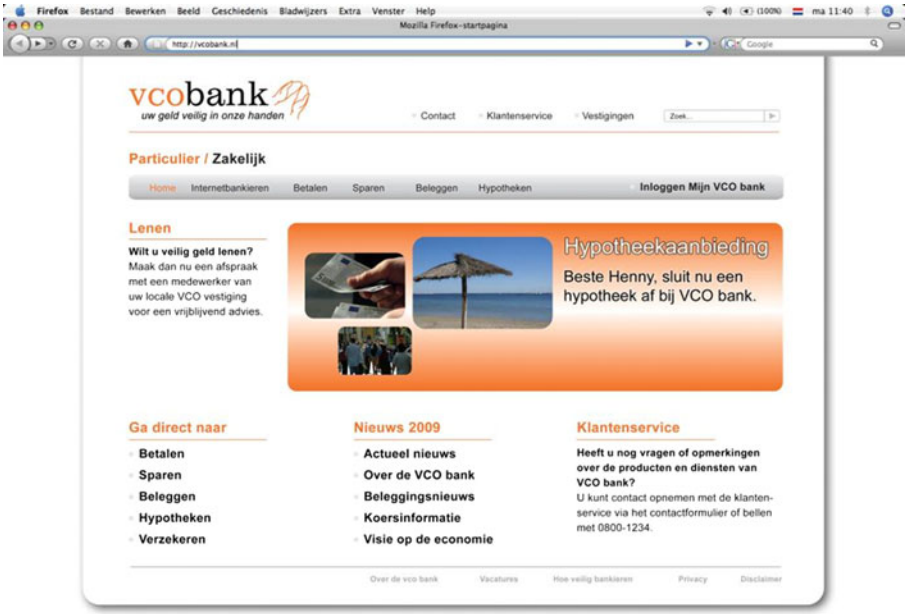
Fig. 2 Impact of personalization and fit of the ad on purchase intentions

In our second study, we also investigated whether the effect of an instrument that is often used to stimulate sales and that offers a potential benefit related to customized advertising, namely, offering a discount, might be attenuated by a sense of intrusiveness. Surprisingly, discounts had no effect on purchase intentions, in contrast with findings in prior promotional literature pertaining to the effectiveness of promotions and discounts (e.g., Bijmolt et al. 2005). A possible explanation is that the intrusiveness that consumers experience might disrupt the interaction with the content (Edwards et al. 2002) and prevent them from taking notice of the benefits of the offer (Morimoto and Chang 2006). Another explanation may be that discounts are so frequent in the telecom industry that our respondents simply do not perceive receiving a discount as something special.

In both studies, we find that respondents with higher levels of privacy concerns experience stronger feelings of intrusiveness, though only in the second study do these concerns affect purchase intentions. We find only a main effect of privacy concern; respondents with higher levels of privacy concerns simply are less likely to purchase. This result might be interpreted in the light of Goldfarb and Tucker's (2011) suggestion that privacy concerns result in a prevention focus that increases consumers' sensitivity to being manipulated by targeted ads. The result may be reactance toward the advertiser (e.g., Clee and Wicklund 1980). Overall, our study clearly and convincingly indicates that customized online advertising is a double-edged sword: It increases purchase intentions, along with feelings of intrusiveness that negatively affect purchase intentions.

Appendix: Examples of stimuli used in the studies

Study 1: Offer with high fit using the person's name



Study 2: Offer with low fit and discount, using transaction information

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